

Seminar Note on FluidWorks UI and Simulation Workflow Debugging

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1. Introduction

FluidWorks is a Unity-based desktop application for CFD-style visualization workflows such as external aerodynamics, internal flow, rotating machinery, and free-surface flow. In a seminar setting, this case study is useful because it demonstrates how software engineering decisions directly affect simulation usability. The work summarized here focuses on model loading, user-interface routing, workflow panels, and rendering reliability.

2. Objective

The objective of this seminar note is to explain how a set of connected usability and rendering issues were identified and corrected in FluidWorks. The intended outcomes were:

- keep the **Model** tab always available,
- open the correct properties panel when a user clicks an item in the simulation outline,
- remove the unnecessary second CAD popup during model import,
- ensure **Simulation Context** is the default panel when a workflow opens,
- keep internal-flow models visible after inlet and outlet assignment,
- restore visible pressure-color feedback in the external aero workflow.

3. Problem Statement

During testing, several linked issues were observed:

- the properties panel routing was inconsistent, so **Simulation Context** sometimes opened the wrong panel,
- **Fluid Properties** was wired to the wrong controller and did not show the expected fluid controls,
- internal and rotatory imports still triggered a second CAD-file popup,
- some tabs were being disabled in code, which made the UI feel broken,
- internal-flow models could appear invisible after assigning inlet and outlet conditions,
- pressure variation colors in the external aero mode were not being shown correctly.

4. Proposed Solution

The proposed solution was implemented through focused corrections in the user-interface controller, runtime model loader, and internal rendering path:

- remove all tab-button disabling so the **Model** tab remains clickable at all times,
- correct outline-to-panel routing in the main UI controller,
- bind **Fluid Properties** to its own dedicated controller instead of reusing wind-tunnel controls,
- force one-file import by disabling the simulation-CAD prompt both in the loader and in template configuration,
- make each workflow open directly on its **Simulation Context** panel,
- auto-open the properties sidebar when a user selects an outline item,
- delay internal heatmap-material replacement until valid solver values exist, so the original model stays visible during setup,
- route external **Pressure** visualization through the pressure render path so pressure colors are visible again.

5. Screenshots of the Solution

Screenshot Placeholder

This seminar PDF was generated without direct filesystem paths to the screenshots.

Recommended screenshots to place here:

- External Aero with **Simulation Context** open by default,
- Internal Flow with the **Model** tab visible and clickable,
- Single-file import dialog showing only the model-file prompt,
- Internal flow after inlet/outlet assignment with the model still visible,
- External aero pressure mode showing visible pressure color variation.

6. Conclusion

This seminar case shows that simulation software quality depends not only on solver correctness, but also on workflow clarity, panel routing, and rendering stability. A model may technically load, yet the system still feels broken if the wrong panel opens, if an unnecessary file popup interrupts the process, or if the object becomes invisible during configuration. The main lesson is to isolate problems by category: interface routing, import workflow, material state, and solver visualization. Once separated, each issue becomes easier to diagnose and fix in a disciplined engineering manner.

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